GOTHAM: iterative Wiener-system identification

Start

Select model parametrization

$$\mathcal{W}_{\mathrm{PL}}^{(k)}(\theta) = \widehat{\Phi}^{(k)}\theta \quad \text{or} \quad \mathcal{W}_{\mathrm{FSM}}(\theta)$$

Select matching loss functional

$$J_{\mathrm{PL}}^{(k)}(\theta) = \tfrac{1}{2}\|C - \widehat{\Phi}^{(k)}\theta\|_2^2$$

$$J_{\nu,\mathrm{FSM}}(\theta) = \tfrac{1}{2}\sum_t r_{\nu,\mathrm{FSM}}(t;\theta)^2$$

Choose inverse-problem formulation

gradient-root or loss-root

$$\mathcal{N}_{\mathrm{G}}(\theta) = \nabla_\theta J(\theta) \quad \text{or} \quad \mathcal{N}_{\mathrm{L}}(\theta) = J(\theta) - \varepsilon_0$$

Set up the zero-order deformation equation

$$(1-q)\,\mathcal{L}(\phi(q) - \theta_0) = q\,\hbar\,H\,\mathcal{N}(\phi(q))$$

Choose $\mathcal{L}, \mathcal{L}^\dagger, H, \hbar, \hat{\theta}^{(0)}, M, K, \delta_{\mathrm{stop}}$

$$k \leftarrow 0, \quad \theta_0^{(k)} \leftarrow \hat{\theta}^{(k)}, \quad m \leftarrow 1$$

$k \geq K$ or $\|N(\hat{\theta}^{(k)})\| \leq \delta_{\mathrm{stop}}$?

yes → Return $\hat{\theta}^{(k)}$

no

$m > M$?

yes → Update parameter estimate

$$\hat{\theta}^{(k+1)} \leftarrow \sum_{j=0}^{M} \theta_j^{(k)}$$

$$k \leftarrow k+1, \quad \theta_0^{(k)} \leftarrow \hat{\theta}^{(k)}, \quad m \leftarrow 1$$

Outer loop

no

Solve the m-th HODE

$$R_m^{(k)} = \mathscr{D}_{m-1}\left(\mathcal{N}\left(\phi^{(k)}\right)\right)$$

$$\theta_m^{(k)} = \chi_m \theta_{m-1}^{(k)} + \mathcal{L}^\dagger\left(\hbar H R_m^{(k)}\right)$$

$m \leftarrow m+1$

Inner loop